

The Common-Reconstruction Rate–Distortion Function with Noisy Context

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Abstract

An encoder observes a jointly Gaussian pair (Y, V) and describes Y at rate R for a consumer that sees the description alone. A third party holds a noisy copy $S = V + U$ of the context V . Neither the consumer nor the third party decodes with S ; instead S prices the description's conditional content $L = I(X; \hat{X} | S)$, the minimal ideal work, in units of $k_B T \ln 2$, that a reset mechanism retaining S must pay to erase the record. We characterize the minimal conditional content at consumer distortion D . The function has a closed form: $L(D) = \frac{1}{2} \log_2 g^*$, where g^* is the larger root of an explicit quadratic in the correlation ρ between read and context and the context noise τ^2 , in any ambient dimension. It recovers the classical rate–distortion function, Gray's conditional rate–distortion function, and Steinberg's common-reconstruction function as degenerations, and it is not a functional of the (Y, S) joint law alone: two instances with the same (Y, S) law and different encoder access have different L . We give the exact rate–content region, whose Pareto frontier is a two-water-level system, and prove that a strict tradeoff between rate and content opens exactly when the context is misaligned with the read. We settle when the natural determinant floor is attained. For the doubly symmetric binary source with a binary symmetric context channel the function closes completely, with an elementary proof, through a tilt equation anchored at Gray's conditional function on one end and the marginal rate–distortion function on the other. Every closed form is verified by independent numerical certificates.

Index Terms

Rate–distortion theory, side information, common reconstruction, conditional mutual information, Landauer principle, Gaussian sources.

I. INTRODUCTION

A stored description of data has two prices. The first is its rate, the space the description occupies, and rate–distortion theory prices it against the fidelity a consumer requires. The second is paid at the end of the description's life: erasing a record irreversibly costs work, and when the mechanism performing the reset retains correlated information, the work is governed not by the record's entropy but by its conditional entropy given what the mechanism holds. The two prices are set by different parties. The consumer never sees the reset mechanism's information, and the reset mechanism never serves the consumer. This paper solves the source coding problem that this asymmetry creates, for Gaussian and binary sources, in closed form.

The setting is as follows. An encoder observes a jointly Gaussian pair (Y, V) : Y is the variable a consumer cares about, V is context correlated with it ($\rho = \mathbb{E}[YV]$ after normalization). The encoder emits a description under a distortion constraint $\mathbb{E}(Y - \hat{Y})^2 \leq D$ that binds the unconditional error; the consumer decodes from the description alone. A third party holds $S = V + U$, a noisy observation of the context with noise variance τ^2 , and possibly a stale one. The quantity of interest is $L(D) = \min I(X; \hat{X} | S)$ over test channels satisfying the distortion constraint and the Markov chain $\hat{X} - X - S$: the minimal conditional content of any admissible description, and, by the operational theorem of Section II, the minimal ideal erasure work per symbol in units of $k_B T \ln 2$.

Three classical objects sit at the corners of this problem, and the function we characterize is none of them. When the read and the context decouple ($\rho = 0$), L is Shannon's rate–distortion function. When the third party's copy is clean ($\tau^2 \rightarrow 0$), L is Gray's conditional rate–distortion function [3]. When read and context merge ($\rho^2 \rightarrow 1$),

L is Steinberg's common-reconstruction function [7] at the induced correlation. Between the corners the function is new, and the distinction is not cosmetic: we exhibit two instances with the same (Y, S) joint law whose values of L differ (Corollary 16). The encoder's access to the clean context V , which neither the consumer nor the third party shares, is load-bearing. To our knowledge no vector-Gaussian common-reconstruction-type function has been published under any formulation; the scalar Gaussian case is due to Steinberg [7] and, for general jointly Gaussian side information, Lapidoth, Malär, and Wigger [8].

The main results are these. Theorem 6 reduces the problem in any ambient dimension to the pair (ρ, τ^2) . Theorem 10 gives the closed form $L(D) = \frac{1}{2} \log_2 g^*$ with g^* the larger root of an explicit quadratic, together with the achieving channel. Theorem 12 characterizes the exact region of achievable rate–content pairs; its Pareto frontier is traced by a two-water-level stationarity system, and Corollary 15 shows the frontier is nondegenerate, so that rate and content genuinely trade off, exactly when $\rho^2 \in (0, 1)$ and $\tau^2 > 0$. Theorem 17 settles when the natural determinant floor is attained, correcting a plausible conjecture: the attainment condition involves the encoder-accessible conditional covariance, not the third party's. Theorem 18 extends the frontier to vector context. Theorem 19 closes the binary case completely: for a doubly symmetric binary source with a binary symmetric context channel, an elementary and self-contained proof yields the function through a tilt equation, anchored at Gray's binary conditional function at one end and the marginal rate–distortion function at the other.

II. PROBLEM FORMULATION AND THE OPERATIONAL THEOREM

All logarithms are base two. Random variables have finite alphabets unless a Gaussian specialization is stated. Let $(X, S) \sim p_{XS}$, and let (X^n, S^n) be i.i.d. copies. The encoder observes X^n but not S^n . The consumer and its decoder also do not observe S^n . The reset mechanism alone has access to S^n , which remains unchanged by the reset.

A distortion measure is a bounded function

$$d : \mathcal{X} \times \hat{\mathcal{X}} \longrightarrow [0, d_{\max}]. \quad (1)$$

An $(n, \mathcal{M}_n)$ observation code consists of an encoder and decoder

$$f_n : \mathcal{X}^n \rightarrow \mathcal{M}_n, \quad g_n : \mathcal{M}_n \rightarrow \hat{\mathcal{X}}^n. \quad (2)$$

We write $M = f_n(X^n)$ and $\hat{X}^n = g_n(M)$. Randomized encoders do not improve the region below, but allowing them in the converse is harmless.

The three operational quantities are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \quad (3)$$

$$D_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E} d(X_i, \hat{X}_i), \quad (4)$$

$$L_n = \frac{1}{n} H(M | S^n). \quad (5)$$

The dimensionless quantity L_n is the *Landauer content* of the memory relative to the retained side information. Under an ideal conditional reset, the average work satisfies

$$W_n \geq nk_B T \ln 2 L_n. \quad (6)$$

In the classical asymptotic regime, conditional erasure protocols make this bound tight up to sublinear terms [14], [15]. We therefore state the coding theorem in bits per source symbol and restore physical units by multiplying by $k_B T \ln 2$. The bound (6) prices only the logically irreversible reset of the volatile index M ; it does not count the work of acquiring X^n , driving a channel, correcting errors, or finishing in finite time [12], [13]. The codebook, wiring, and any permanent lookup tables are treated as part of the apparatus rather than a per-cycle record.

Definition 1 (Achievability). *A triple (R, L, D) is achievable if there is a sequence of observation codes such that*

$$\limsup_{n \rightarrow \infty} R_n \leq R, \quad \limsup_{n \rightarrow \infty} L_n \leq L, \quad \limsup_{n \rightarrow \infty} D_n \leq D. \quad (7)$$

The closure of the achievable set at distortion D is denoted $\mathcal{RW}(D)$.

For a test channel $p(\hat{x} | x)$, the Markov chain $\hat{X} - X - S$ holds because side information is unavailable when the observation is formed. Define

$$\mathcal{T}(D) = \left\{ p(\hat{x} | x) : \mathbb{E}d(X, \hat{X}) \leq D \right\}. \quad (8)$$

Lemma 2 (Convexity in the test channel). *For fixed p_{XS} , both coordinates $I(X; \hat{X})$ and $I(X; \hat{X} | S)$ are convex functions of the test channel $q(\hat{x} | x)$.*

Proof. Convexity of $I(X; \hat{X})$ in the channel at fixed input is classical. For the conditional coordinate, the Markov chain $\hat{X} - X - S$ gives $p(\hat{x} | x, s) = q(\hat{x} | x)$, so

$$I(X; \hat{X} | S) = \sum_s p(s) I_{p(x|s)}(q), \quad (9)$$

a nonnegative combination of mutual informations, each with fixed input distribution $p(x | s)$ and the common channel q ; each term is convex in q , hence so is the sum. We note that the decomposition $I(X; \hat{X} | S) = I(X; \hat{X}) - I(S; \hat{X})$ does *not* yield this conclusion: $I(S; \hat{X})$ is itself convex in q (the induced channel $p(\hat{x} | s) = \sum_x p(x | s)q(\hat{x} | x)$ is linear in q), so the difference is a priori indefinite. $\square$

Theorem 3 (Rate–work–distortion region). *For a discrete memoryless source (X, S) and bounded distortion d ,*

$$\mathcal{RW}(D) = \bigcup_{p(\hat{x}|x) \in \mathcal{T}(D)} \left\{ (R, L) : R \geq I(X; \hat{X}), L \geq I(X; \hat{X} | S) \right\}. \quad (10)$$

The physical reset-work coordinate is $W/n = k_B T \ln 2 L$ in the ideal model. By Lemma 2 the union is already convex, and it is closed by compactness of $\mathcal{T}(D)$ and continuity of both coordinates; no time sharing is required, and every boundary point is attained by a single test channel.

Proof. We first prove the converse. Let a code produce M and $\hat{X}^n$. Since $\hat{X}^n$ is a function of M ,

$$\begin{aligned} \log |\mathcal{M}_n| &\geq H(M) \geq I(X^n; M) \geq I(X^n; \hat{X}^n) \\ &\geq \sum_{i=1}^n I(X_i; \hat{X}_i), \end{aligned} \quad (11)$$

where the last step follows from memorylessness and conditioning reducing entropy (Appendix A). Likewise,

$$\begin{aligned} H(M | S^n) &\geq I(X^n; M | S^n) \geq I(X^n; \hat{X}^n | S^n) \\ &\geq \sum_{i=1}^n I(X_i; \hat{X}_i | S_i). \end{aligned} \quad (12)$$

Each induced per-letter channel $p(\hat{x}_i | x_i)$ satisfies the Markov constraint $\hat{X}_i - X_i - S_i$: $\hat{X}_i$ is a function of $(X_i, X_{\neq i})$ and S_i is independent of $X_{\neq i}$ given X_i for i.i.d. pairs. Let $\bar{q}(\hat{x} | x) = \frac{1}{n} \sum_{i=1}^n p(\hat{x}_i = \hat{x} | X_i = x)$ be the mixture channel; the constraint survives the mixture because $p(s | x)$ is common across i . Its distortion is D_n , so $\bar{q} \in \mathcal{T}(D_n)$, and by Lemma 2,

$$R_n \geq \frac{1}{n} \sum_i I(X_i; \hat{X}_i) \geq I_{\bar{q}}(X; \hat{X}), \quad L_n \geq \frac{1}{n} \sum_i I(X_i; \hat{X}_i | S_i) \geq I_{\bar{q}}(X; \hat{X} | S),$$

which places (R_n, L_n) in the (already convex and closed) union of (10) at distortion D_n .

For achievability, fix $p(\hat{x} | x) \in \mathcal{T}(D)$ and $\epsilon > 0$. Generate a standard rate–distortion codebook containing $2^{n(I(X; \hat{X}) + 2\epsilon)}$ independent $\hat{x}^n$ codewords. A covering encoder selects a codeword jointly typical with x^n and stores its index M . The consumer reconstructs that codeword from M alone. Standard typicality arguments give rate $I(X; \hat{X}) + 2\epsilon$ and distortion at most $D + o(1)$.

To bound the reset work, randomly partition the codeword indices into $2^{n(I(X; \hat{X}|S) + 3\epsilon)}$ bins, and let $B = b(M)$ be the bin index. Because $\hat{X} - X - S$,

$$I(X; \hat{X} | S) = I(X; \hat{X}) - I(S; \hat{X}). \quad (13)$$

The usual Wyner–Ziv packing argument [4] therefore gives a decoder that recovers M from (B, S^n) with probability of error tending to zero. This decoder is a proof device; the consumer still receives no side information. By Fano’s inequality,

$$\begin{aligned} H(M | S^n) &\leq H(B) + H(M | B, S^n) \\ &\leq n(I(X; \hat{X} | S) + 3\epsilon) + o(n). \end{aligned} \quad (14)$$

Thus the same stored index has ordinary description rate $I(X; \hat{X})$ and conditional Landauer content $I(X; \hat{X} | S)$. Since the union in (10) is already convex and closed (Lemma 2), this exhibits the entire region. $\square$

The binning step in the proof has a simple interpretation. The full codeword index is required by the consumer. At reset, side information identifies the codeword up to a bin of asymptotic size $2^{nI(X; \hat{X} | S)}$. Only that residual uncertainty is thermodynamically irreducible. The step has also been verified numerically at finite blocklength, by in-bin decoders that succeed from (B, S^n) near the predicted bin rate and fail at every bin rate without S^n .

The work-only endpoint of the region is the minimum Landauer content

$$L(D) = \min_{p(\hat{x}|x) \in \mathcal{T}(D)} I(X; \hat{X} | S). \quad (15)$$

As an information quantity, $L(D)$ is Steinberg’s common-reconstruction rate–distortion function for side information S [7]: the reproduction must be determined without access to S . The operational roles differ. In the common-reconstruction problem, S is decoder side information that must not corrupt reproducibility; here S is held by a reset mechanism that never serves the consumer. The remainder of the paper characterizes $L(D)$, and the region around it, for the source of Section III.

Remark 4 (The Gaussian pair specialization). *From Section III on, the source letter is the jointly Gaussian pair $X = (Y, V)$ introduced there, and the distortion is $d(x, \hat{x}) = (y - \hat{y})^2$ on the read coordinate: the consumer is judged on Y alone, while the encoder observes the pair.*

Remark 5 (Scope of the coding theorem). *Theorem 3 is stated and proved for finite alphabets and bounded per-letter distortion. Its application to the Gaussian source with unbounded quadratic distortion is the standard abstract-alphabet extension of the coding theorem [2]. The Gaussian closed forms of Sections III–VI are evaluations of the single-letter coordinates in (10), and their operational meaning is inherited through that extension.*

III. PAIR SUFFICIENCY

Let $X \sim \mathcal{N}(0, \Sigma_x)$ on $\mathbb{R}^d$, i.i.d. across source symbols. The consumer’s read is the linear functional $Y = w^\top X$; the third party’s copy is a second linear functional observed in noise, $S = \gamma^\top X + U$. Normalize $\text{Var}(w^\top X) = 1$ (rescaling D), write $V = \gamma^\top X / \sigma_\gamma$ with $\rho := \mathbb{E}[YV]$, $|\rho| < 1$, and rescale S (invariantly for all mutual informations) to $S = V + U$, $U \sim \mathcal{N}(0, \tau^2)$, $\tau^2 > 0$. Write $s = 1 + \tau^2$. Throughout, X , U , and all channel randomness (including any resampling randomness below) are *mutually* independent. This joint form of the independence assumption is load-bearing: it is exactly what the S -kernel argument in the next proof uses.

Reproductions may be taken scalar without loss of optimality: replacing $\hat{X}$ by $\hat{Y} := w^\top \hat{X}$ leaves the distortion unchanged and can only decrease $I(X; \cdot | S)$ (conditional data processing), and $\hat{Y} - X - S$ persists.

Theorem 6 (Pair sufficiency). *In the single-consumer problem $L(D) = \min\{I(X; \hat{Y} | S) : \hat{Y} - X - S, \mathbb{E}(Y - \hat{Y})^2 \leq D\}$, it is without loss of optimality to let the channel depend on X only through $T = (Y, V)$, and then $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. Consequently $L(D)$ depends on $(\Sigma_x, w, \gamma, \tau^2)$, in any ambient dimension d , only through (ρ, τ^2) .*

Proof. Given an admissible $p(\hat{y} | x)$, draw $\hat{Y}'$ from the conditional law $p(\hat{y} | T = t)$ using fresh randomness. (a) $(T, \hat{Y}') \stackrel{d}{=} (T, \hat{Y})$, so the distortion (a functional of $(Y, \hat{Y})$) is unchanged. (b) $S = V + U$ with U independent of $(X, \hat{Y})$ and of $(X, \hat{Y}')$, so in both joints $p(s | t, \hat{y}) = p_U(s - v)$: hence $(T, \hat{Y}', S) \stackrel{d}{=} (T, \hat{Y}, S)$. (c) Given T , $\hat{Y}'$ is independent of (X, S) , so $I(X; \hat{Y}' | S) = I(T; \hat{Y}' | S) + I(X; \hat{Y}' | T, S) = I(T; \hat{Y} | S) \leq I(X; \hat{Y} | S)$, the last step because T is a function of X . (d) $\hat{Y}' - X - S$ holds since $\hat{Y}'$ is a function of $(X, \text{fresh noise})$ with the noise independent of (X, U) . The same computation applied to a T -channel directly gives $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. $\square$

IV. THE FUNCTION: SCALAR CONTEXT

One further reduction identifies the class the closed form must beat. (*Marginalized class = scalar corner.*) If the channel depends on X only through Y (that is, $\hat{Y} - Y - (X, S)$), then given (Y, S) , $\hat{Y}$ is independent of X , so $I(X; \hat{Y} | S) = I(Y; \hat{Y} | S)$, and the class optimum is exactly the scalar-corner value $\frac{1}{2} \log_2 [(\sigma^2(1 - \rho_{YS}^2) + \rho_{YS}^2 D)/D]$ of the pair (Y, S) [16], [7]; the corner's converse applies to every $p(\hat{y} | y)$, discrete or not.

Proposition 7 (Marginalization is strictly suboptimal). *Let $d = 2$, $X \sim \mathcal{N}(0, I_2)$, $Y = X_1$, and $S = X_1 + X_2$ (so $\rho_{YS}^2 = \frac{1}{2}$ and S is X -measurable). For $0 < D < 1$:*

- (i) *the marginalized-class optimum is the scalar corner $L_{\text{marg}}(D) = \frac{1}{2} \log_2 \frac{1+D}{2D}$;*
- (ii) *the vector problem has the exact value*

$$L_{\text{vec}}(D) = \frac{1}{2} \log_2^+ \frac{1}{2D} = R_{Y|S}(D),$$

Gray's conditional rate-distortion function of (Y, S) [3] ($\text{Var}(Y | S) = \frac{1}{2}$), attained by the linear channel $\hat{Y} = (1 - D)X_1 + DX_2 + N$, $N \sim \mathcal{N}(0, D(1 - 2D))$, for $D \leq \frac{1}{2}$, and by $\hat{Y} = S/2$ (zero conditional content) for $D \geq \frac{1}{2}$;

- (iii) *hence the strict gap $L_{\text{marg}}(D) - L_{\text{vec}}(D) = \frac{1}{2} \log_2(1 + D) > 0$ on $(0, \frac{1}{2}]$, and equal to the full marginalized value on $[\frac{1}{2}, 1)$ (where $L_{\text{vec}} = 0$; the absolute gap is decreasing there, but it is all of L_{marg}).*

Proof. (i) is the reduction above, evaluated at $\sigma^2 = 1$, $\rho_{YS}^2 = \frac{1}{2}$.

(ii) *Converse.* For any admissible $\hat{Y}$ (Markov $\hat{Y} - X - S$), $I(X; \hat{Y} | S) \geq I(Y; \hat{Y} | S)$, and the induced conditional channel $p(\hat{y} | y, s)$ is feasible for Gray's conditional problem at the same distortion, so $I(Y; \hat{Y} | S) \geq R_{Y|S}(D) = \frac{1}{2} \log_2^+(\text{Var}(Y | S)/D) = \frac{1}{2} \log_2^+(1/(2D))$. *Achievability.* For $D \leq \frac{1}{2}$ take $\hat{Y} = (1 - D)X_1 + DX_2 + N$: distortion $D^2 + D^2 + D(1 - 2D) = D$; $\text{Var}(\hat{Y}) = 1 - D$, $\text{Cov}(\hat{Y}, S) = 1$, $\text{Var}(\hat{Y} | S) = 1 - D - \frac{1}{2} = \frac{1}{2} - D$, so $I(X; \hat{Y} | S) = \frac{1}{2} \log_2 \frac{(1-2D)/2}{D(1-2D)} = \frac{1}{2} \log_2 \frac{1}{2D}$. For $D \geq \frac{1}{2}$, $\hat{Y} = S/2$ is a deterministic function of X , has distortion $\mathbb{E}((X_1 - X_2)/2)^2 = \frac{1}{2} \leq D$, and is constant given S , so $I(X; \hat{Y} | S) = 0$.

(iii) Subtract. The closed forms on both sides have been verified numerically. $\square$

The closed form occupies the rest of this section. By Theorem 6 we restrict to T -channels, $T = (Y, V)$, with scalar reproduction $\hat{Y}$. The converse rests on one estimation-theoretic lemma. We state it for vector reproductions and vector conditioners, in the generality that Sections V, VI, and VII reuse.

Lemma 8 (Gaussian exhaustion and the determinant identity). *Let (T, S) be jointly Gaussian and centered, $T \in \mathbb{R}^p$, $S \in \mathbb{R}^r$, with $\text{Cov}((T, S)) \succ 0$. Let $\hat{\mathbf{Y}} \in \mathbb{R}^m$ be any centered random vector with finite second moments, and define*

$$A := \text{Cov}(\hat{\mathbf{Y}}, T) \Sigma_T^{-1} \in \mathbb{R}^{m \times p}, \quad N' := \hat{\mathbf{Y}} - AT, \quad \Sigma_N := \text{Cov}(N') = \text{Cov}(\hat{\mathbf{Y}}) - A \Sigma_T A^\top.$$

Assume $\Sigma_N \succ 0$ and $\text{Cov}(N', S) = 0$. Then

$$I(T; \hat{\mathbf{Y}} | S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{1}{2} \log_2 \frac{\det(A \Sigma_{T|S} A^\top + \Sigma_N)}{\det \Sigma_N}, \quad (16)$$

where $\Sigma_{T|S} = \Sigma_T - \Sigma_{TS} \Sigma_S^{-1} \Sigma_{ST}$ and Σ_e is the error covariance of the best linear estimator of T from $(\hat{\mathbf{Y}}, S)$. With S deleted throughout, the hypothesis $\text{Cov}(N', S) = 0$ becomes vacuous and the same argument gives

$$I(T; \hat{\mathbf{Y}}) \geq \frac{1}{2} \log_2 \frac{\det(A \Sigma_T A^\top + \Sigma_N)}{\det \Sigma_N}. \quad (17)$$

The jointly Gaussian channel $\hat{\mathbf{Y}} = AT + N$, $N \sim \mathcal{N}(0, \Sigma_N)$ independent of (T, S) , realizes the same second moments and attains both bounds with equality simultaneously.

Proof. Step 1: the max-entropy bound. Write $M := \text{Cov}((\hat{\mathbf{Y}}, S))$ and $J := \text{Cov}(T, (\hat{\mathbf{Y}}, S)) \in \mathbb{R}^{p \times (m+r)}$. First, $M \succ 0$: by the Schur-complement criterion on

$$M = \begin{bmatrix} \text{Cov}(\hat{\mathbf{Y}}) & \text{Cov}(\hat{\mathbf{Y}}, S) \\ \text{Cov}(S, \hat{\mathbf{Y}}) & \Sigma_S \end{bmatrix},$$

positivity is equivalent to $\Sigma_S \succ 0$ together with $\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S)\Sigma_S^{-1}\text{Cov}(S, \hat{\mathbf{Y}}) \succ 0$; the first holds because $\text{Cov}((T, S)) \succ 0$, and the second is evaluated in Step 2 to equal $A\Sigma_{T|S}A^\top + \Sigma_N \succeq \Sigma_N \succ 0$. The best linear estimator of T from $(\hat{\mathbf{Y}}, S)$ is $\hat{T} = JM^{-1}(\hat{\mathbf{Y}}; S)$, with error covariance $\Sigma_e = \Sigma_T - JM^{-1}J^\top$. Since (T, S) is jointly Gaussian, $h(T | S) = \frac{1}{2} \log_2((2\pi e)^p \det \Sigma_{T|S})$ exactly. For the other term, translation invariance of differential entropy, conditioning reduces entropy, and the Gaussian maximum-entropy bound at fixed covariance give

$$h(T | \hat{\mathbf{Y}}, S) = h(T - \hat{T} | \hat{\mathbf{Y}}, S) \leq h(T - \hat{T}) \leq \frac{1}{2} \log_2((2\pi e)^p \det \Sigma_e)$$

(with the usual convention that $h = -\infty$ for laws without a density, in which case the bound is trivial). Subtracting the two displays inside $I(T; \hat{\mathbf{Y}} | S) = h(T | S) - h(T | \hat{\mathbf{Y}}, S)$ yields the inequality in (16).

Step 2: the determinant identity. Let $K := \text{Cov}((T, S, \hat{\mathbf{Y}}))$. The residual N' is uncorrelated with T by construction of the regression and with S by hypothesis, and the change of variables $(T, S, \hat{\mathbf{Y}}) \mapsto (T, S, N')$ is block lower triangular with unit diagonal, hence determinant preserving:

$$\det K = \det \text{Cov}((T, S, N')) = \det \begin{bmatrix} \Sigma_T & \Sigma_{TS} & 0 \\ \Sigma_{ST} & \Sigma_S & 0 \\ 0 & 0 & \Sigma_N \end{bmatrix} = \det \Sigma_S \det \Sigma_{T|S} \det \Sigma_N,$$

using in the last step the Schur factorization $\det \text{Cov}((T, S)) = \det \Sigma_S \det \Sigma_{T|S}$. The same Schur factorization applied to K in the other block order gives $\det K = \det M \det \Sigma_e$, and applied to M itself gives

$$\det M = \det \Sigma_S \cdot \det \left(\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S)\Sigma_S^{-1}\text{Cov}(S, \hat{\mathbf{Y}}) \right).$$

It remains to evaluate the conditional block. From $\hat{\mathbf{Y}} = AT + N'$ with $\text{Cov}(N', T) = 0$ and $\text{Cov}(N', S) = 0$, $\text{Cov}(\hat{\mathbf{Y}}) = A\Sigma_T A^\top + \Sigma_N$ and $\text{Cov}(\hat{\mathbf{Y}}, S) = A\Sigma_{TS}$, so

$$\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S)\Sigma_S^{-1}\text{Cov}(S, \hat{\mathbf{Y}}) = A(\Sigma_T - \Sigma_{TS}\Sigma_S^{-1}\Sigma_{ST})A^\top + \Sigma_N = A\Sigma_{T|S}A^\top + \Sigma_N.$$

Combining the three displays,

$$\det \Sigma_e = \frac{\det K}{\det M} = \frac{\det \Sigma_{T|S} \det \Sigma_N}{\det (A\Sigma_{T|S}A^\top + \Sigma_N)},$$

which is the identity in (16); in particular $\Sigma_e \succ 0$, so Step 1's bound is finite. Deleting S (formally $r = 0$: $K = \text{Cov}((T, \hat{\mathbf{Y}}))$, $M = \text{Cov}(\hat{\mathbf{Y}})$) gives $\det \Sigma_{e0} = \det \Sigma_T \det \Sigma_N / \det (A\Sigma_T A^\top + \Sigma_N)$ and hence (17).

Step 3: equality. For the stated Gaussian channel the conditional law of T given $(\hat{\mathbf{Y}}, S)$ is Gaussian with covariance Σ_e for every realization, so $h(T | \hat{\mathbf{Y}}, S) = \frac{1}{2} \log_2((2\pi e)^p \det \Sigma_e)$ and (16) holds with equality; likewise for (17). The channel's moments match those of the given channel: $\text{Cov}(AT + N, T) = A\Sigma_T$ and $\text{Cov}(AT + N) = A\Sigma_T A^\top + \Sigma_N$. $\square$

Remark 9 (Degenerate residuals). If Σ_N is singular, some combination $\eta^\top \hat{\mathbf{Y}}$ equals $\eta^\top AT$ almost surely. If $\eta^\top A \neq 0$ the record reveals a noiseless linear functional of the continuous T , so $I(T; \hat{\mathbf{Y}} | S) = \infty$ and every bound below holds trivially; if $\eta^\top A = 0$ that combination of the record is zero and may be dropped, reducing m . All uses of Lemma 8 below may therefore assume $\Sigma_N \succ 0$.

Theorem 10 (The function). Write $s = 1 + \tau^2$. For $0 < D < 1$,

$$L(D) = \frac{1}{2} \log_2 g^*, \quad g^* = \frac{(D + s - \rho^2) + \sqrt{(D + s - \rho^2)^2 - 4Ds(1 - \rho^2)}}{2Ds}, \quad (18)$$

the larger root of the quadratic

$$P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2). \quad (19)$$

Since $P(1) = (D - 1)\tau^2 < 0$, exactly one root exceeds 1, so g^* is well defined. The optimum is attained by the jointly Gaussian channel

$$\hat{Y} = aY + bV + N, \quad a = \frac{g^* - 1}{g^*}, \quad b = \frac{(g^* - 1)\rho}{g^*k}, \quad k = g^*s - 1, \quad N \sim \mathcal{N}\left(0, \frac{Q_1}{g^* - 1}\right), \quad (20)$$

where $Q_1 = a^2 + b^2 + 2ab\rho - (a\rho + b)^2/s$ is evaluated at these (a, b) .

Proof. Moments. By Theorem 6 and the scalar-reproduction reduction of Section III, restrict to channels $p(\hat{y} | t)$. Fix an admissible channel with $\mathbb{E}(Y - \hat{Y})^2 \leq D$. Centering $\hat{Y}$ leaves $I(T; \hat{Y} | S)$ unchanged and can only decrease the distortion, so take $\mathbb{E}\hat{Y} = 0$. Define the regression coordinates

$$\begin{pmatrix} a \\ b \end{pmatrix} := \Sigma_T^{-1} \begin{pmatrix} \mathbb{E}[Y\hat{Y}] \\ \mathbb{E}[V\hat{Y}] \end{pmatrix}, \quad Q_0 := a^2 + b^2 + 2ab\rho = \text{Var}(aY + bV), \quad n := \text{Var}\hat{Y} - Q_0,$$

so that $\mathbb{E}[Y\hat{Y}] = a + b\rho$ and $\mathbb{E}[V\hat{Y}] = a\rho + b$; $n \geq 0$ is the residual variance of $\hat{Y}$ after linear regression on T . If $n = 0$, Remark 9 applies: either $(a, b) \neq (0, 0)$ and $I(T; \hat{Y} | S) = \infty$, or $(a, b) = (0, 0)$ and $\hat{Y} = 0$ with distortion $1 > D$. So take $n > 0$.

The Markov moment identity. The residual $N' = \hat{Y} - aY - bV$ is uncorrelated with T by construction. It is also uncorrelated with $S = V + U$: it is uncorrelated with V , a coordinate of T , and $\mathbb{E}[UN'] = 0$ because U is independent of $(X, \hat{Y})$ by the mutual independence of X , U , and the channel randomness. Hence $\text{Cov}(N', S) = 0$, which is the hypothesis of Lemma 8, and which restates the moment identity $\mathbb{E}[S\hat{Y}] = \mathbb{E}[V\hat{Y}]$.

The converse bound, written out. Here $p = 2$, $m = 1$, $r = 1$, $A = (a, b)$, $\Sigma_N = n$, and

$$\Sigma_T = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \quad \Sigma_{TS} = \begin{bmatrix} \rho \\ 1 \end{bmatrix}, \quad \Sigma_S = s,$$

$$\Sigma_{T|S} = \Sigma_T - \frac{1}{s} \Sigma_{TS} \Sigma_{TS}^\top = \begin{bmatrix} 1 - \rho^2/s & \rho\tau^2/s \\ \rho\tau^2/s & \tau^2/s \end{bmatrix}, \quad \det \Sigma_{T|S} = \frac{\tau^2(1 - \rho^2)}{s} > 0.$$

The quadratic form in (16) is

$$A \Sigma_{T|S} A^\top = Q_0 - \frac{(a\rho + b)^2}{s} = \text{Var}(aY + bV | S) =: Q_1,$$

and, explicitly, $M = \text{Cov}((\hat{Y}, S)) = \begin{bmatrix} Q_0 + n & a\rho + b \\ a\rho + b & s \end{bmatrix}$ with $\det M = s(Q_0 + n) - (a\rho + b)^2 = s(Q_1 + n)$, while $\det \text{Cov}((T, S, \hat{Y})) = s \det \Sigma_{T|S} \cdot n$, so $\det \Sigma_e = n \det \Sigma_{T|S} / (Q_1 + n)$ and Lemma 8 reads

$$I(T; \hat{Y} | S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{1}{2} \log_2 \frac{Q_1 + n}{n}. \quad (21)$$

The distortion is a function of the same moments:

$$\mathbb{E}(Y - \hat{Y})^2 = 1 - 2(a + b\rho) + Q_0 + n = h(a, b) + n, \quad h(a, b) := (1 - a)^2 - 2(1 - a)b\rho + b^2, \quad (22)$$

and $h(a, b) = \text{Var}((1 - a)Y - bV) \geq 0$. Therefore

$$L(D) \geq \inf \left\{ B(a, b, n) := \frac{1}{2} \log_2 \frac{Q_1 + n}{n} : h(a, b) + n \leq D, n > 0 \right\}, \quad (23)$$

and it suffices to evaluate the right side; achievability below matches it.

Activity, coercivity, existence. At fixed (a, b) with $Q_1 > 0$, B is strictly decreasing in n ($\partial_n [\ln(Q_1 + n) - \ln n] = (Q_1 + n)^{-1} - n^{-1} < 0$), so the infimum is unchanged by imposing the active constraint $n = D - h(a, b)$. On the feasible set, (a, b) is confined to the compact set $\{h \leq D\}$; since $\Sigma_{T|S} \succ 0$, the form $Q_1 = (a, b) \Sigma_{T|S} (a, b)^\top$ vanishes only at $(a, b) = (0, 0)$, where $h = 1 > D$, so by continuity $Q_1 \geq q_0 > 0$ on the feasible set. Hence $B \geq \frac{1}{2} \log_2 ((q_0 + n)/n) \rightarrow \infty$ as $n \rightarrow 0$, the sublevel sets of the reduced objective

$$\Phi(a, b) := \frac{1}{2} \log_2 \frac{Q_1(a, b) + n(a, b)}{n(a, b)}, \quad n(a, b) = D - h(a, b),$$

are compact, and the infimum in (23) is attained at some (a, b) with $h(a, b) < D$: an interior stationary point of Φ .

The stationarity system is linear in (a, b) at fixed g . At a stationary point write $g := (Q_1 + n)/n$ and $\mu_c := (a\rho + b)/s$; note $g > 1$ because $Q_1 > 0$. The partial derivatives of the three quadratics are

$$\begin{aligned} \partial_a Q_0 &= 2(a + b\rho), & \partial_b Q_0 &= 2(b + a\rho), & \partial_a h &= \partial_a Q_0 - 2, & \partial_b h &= \partial_b Q_0 - 2\rho, \\ \partial_a Q_1 &= \partial_a Q_0 - 2\rho\mu_c, & \partial_b Q_1 &= \partial_b Q_0 - 2\mu_c, \end{aligned}$$

hence the four gradient identities

$$\partial_a(Q_0 - h) = 2, \quad \partial_b(Q_0 - h) = 2\rho, \quad \partial_a(Q_1 - h) = 2(1 - \rho\mu_c), \quad \partial_b(Q_1 - h) = 2(\rho - \mu_c). \quad (24)$$

(Only the last two enter here; all four enter the weighted system of Theorem 12.) With $n = D - h$, so $\partial_x n = -\partial_x h$, stationarity of $2 \ln 2 \cdot \Phi = \ln(Q_1 + n) - \ln n$ in $x \in \{a, b\}$ reads

$$\frac{\partial_x(Q_1 - h)}{Q_1 + n} + \frac{\partial_x h}{n} = 0, \quad \text{that is,} \quad \frac{1}{g} \partial_x(Q_1 - h) + \partial_x h = 0.$$

By (24) and $\partial_a h = 2(a + b\rho - 1)$, $\partial_b h = 2(a\rho + b - \rho)$, the two equations are

$$a + b\rho = 1 - \frac{1 - \rho\mu_c}{g}, \quad a\rho + b = \rho - \frac{\rho - \mu_c}{g}. \quad (25)$$

Substituting $\mu_c = (a\rho + b)/s$ makes (25) a 2×2 linear system in (a, b) whose coefficients depend on g alone, and it solves in closed form. The second equation reads $gs\mu_c - \mu_c = \rho(g - 1)$, so with $k := gs - 1 > 0$,

$$\mu_c = \frac{\rho(g - 1)}{k}. \quad (26)$$

Subtracting ρ times the first equation of (25) from the second gives $(a\rho + b) - \rho(a + b\rho) = b(1 - \rho^2)$ on the left and $\mu_c(1 - \rho^2)/g$ on the right, so $b = \mu_c/g$; the first equation then yields $a = 1 - (1 - \rho\mu_c)/g - b\rho = 1 - 1/g$. The stationary channel coefficients at level g are therefore

$$1 - a = \frac{1}{g}, \quad b = \frac{\mu_c}{g} = \frac{(g - 1)\rho}{gk}. \quad (27)$$

Reduction to the quadratic. At the values (27), $(1 - a)Y - bV = (Y - \mu_c V)/g$, so

$$h = \frac{1 - 2\rho\mu_c + \mu_c^2}{g^2}, \quad (28)$$

and, writing $aY + bV = Y - (Y - \mu_c V)/g$ and using the definition $a\rho + b = s\mu_c$,

$$Q_0 = 1 - \frac{2(1 - \rho\mu_c)}{g} + h, \quad Q_1 = Q_0 - \frac{(a\rho + b)^2}{s} = 1 - \frac{2(1 - \rho\mu_c)}{g} + h - s\mu_c^2.$$

The definition of g and the active constraint combine to $(g - 1)(D - h) = Q_1$. Substituting the display for Q_1 and collecting the h terms ($h + (g - 1)h = gh = (1 - 2\rho\mu_c + \mu_c^2)/g$ by (28)), the terms in $\rho\mu_c$ cancel and

$$1 + \frac{\mu_c^2 - 1}{g} - s\mu_c^2 = (g - 1)D. \quad (29)$$

Finally substitute (26). Multiplying (29) by gk^2 and using $k^2\mu_c^2 = \rho^2(g - 1)^2$,

$$gk^2 + \rho^2(g - 1)^2 - k^2 - gs\rho^2(g - 1)^2 = g(g - 1)Dk^2.$$

Since $1 - gs = -k$, the second and fourth terms on the left combine to $-\rho^2(g - 1)^2k$, so the left side is $(g - 1)k[k - \rho^2(g - 1)]$, and dividing by $(g - 1)k > 0$,

$$k - \rho^2(g - 1) = gDk \iff Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2) = 0,$$

that is, $P(g) = 0$. Every interior stationary point of Φ therefore sits at a root of P exceeding 1.

Root bracketing and uniqueness. $P(1) = Ds - (D + s - \rho^2) + (1 - \rho^2) = (D - 1)(s - 1) = (D - 1)\tau^2 < 0$ for $D < 1$, $\tau^2 > 0$. The leading coefficient Ds is positive and $P(0) = 1 - \rho^2 > 0$, so P has two real roots, exactly one exceeding 1 (the other lies in $(0, 1)$). Hence every stationary point has $g = g^*$, and by (27) the stationary (a, b) , and with it $n = Q_1/(g^* - 1)$, is unique. The attained minimum of (23) is $\frac{1}{2} \log_2 g^*$.

Achievability. The channel (20) has exactly the moments (27) with residual variance $n = Q_1/(g^* - 1)$. Its distortion is $h + n$, and $(g^* - 1)n = Q_1$ together with $P(g^*) = 0$ is, by the chain of equivalences above run backward, the statement $h + n = D$. The channel is jointly Gaussian with (T, S) , so the equality clause of Lemma 8 gives

$$I(T; \hat{Y} | S) = \frac{1}{2} \log_2 \frac{Q_1 + n}{n} = \frac{1}{2} \log_2 g^*.$$

Hence $L(D) = \frac{1}{2} \log_2 g^*$. $\square$

Corollary 11 (Anchors). (i) At $\rho = 0$, $P(g) = (Dg - 1)(sg - 1)$ and $g^* = 1/D$: $L(D) = \frac{1}{2} \log_2(1/D)$ is the classical rate–distortion function, attained with $b = 0$ by the classical reverse test channel. (ii) As $\tau^2 \rightarrow 0$, $L(D) \rightarrow \frac{1}{2} \log_2^+((1 - \rho^2)/D)$, Gray’s conditional rate–distortion function of the pair (Y, V) [3], matching Proposition 7. (iii) As $\rho^2 \rightarrow 1$, $L(D) \rightarrow \frac{1}{2} \log_2[(D + \tau^2)/(D(1 + \tau^2))]$, Steinberg’s common-reconstruction function at the induced correlation $\rho_{Y_S}^2 = 1/s$ [7].

Proof. (i) $(Dg - 1)(sg - 1) = Dsg^2 - (D + s)g + 1 = P(g)|_{\rho=0}$. The roots are $1/D > 1$ and $1/s < 1$, so $g^* = 1/D$. At this root (20) gives $b = 0$, $a = 1 - D$, $Q_1 = (1 - D)^2$, and $\text{Var } N = (1 - D)^2/(1/D - 1) = D(1 - D)$: the classical reverse channel.

(ii) At $s = 1$, $P(g) = Dg^2 - (D + 1 - \rho^2)g + (1 - \rho^2) = (g - 1)(Dg - (1 - \rho^2))$, with discriminant $(D - (1 - \rho^2))^2$, so the larger root at $s = 1$ is exactly $\max\{1, (1 - \rho^2)/D\}$; the roots of P are continuous in s , and the limit follows. For $D < 1 - \rho^2$ the value is $\frac{1}{2} \log_2(\text{Var}(Y | V)/D)$ with $\text{Var}(Y | V) = 1 - \rho^2$, Gray’s conditional function; for $D \geq 1 - \rho^2$ it is 0.

(iii) At $\rho^2 = 1$ the constant term of P vanishes: $P(g) = g(Dsg - (D + \tau^2))$, with roots 0 and $(D + \tau^2)/(Ds)$; the latter exceeds 1 because $(D + \tau^2) - Ds = \tau^2(1 - D) > 0$. By root continuity $g^* \rightarrow (D + \tau^2)/(Ds)$ as $\rho^2 \rightarrow 1$. Steinberg’s scalar function at unit source variance and side-information correlation $\rho_{Y_S}^2$ is $\frac{1}{2} \log_2[(1 - \rho_{Y_S}^2 + \rho_{Y_S}^2 D)/D]$ [7], [16]; at $\rho_{Y_S}^2 = \rho^2/s \rightarrow 1/s$ this is $\frac{1}{2} \log_2[(\tau^2 + D)/(sD)]$, the same value. $\square$

V. THE RATE–CONTENT REGION AND MISALIGNMENT

The work-only endpoint of Theorem 10 does not come for free in rate. This section characterizes the full tradeoff between the two coordinates of (10) for the Gaussian pair source. Throughout, $\rho^2 \in (0, 1)$, $\tau^2 > 0$, $0 < D < 1$, and h, Q_0, Q_1, μ_c are as in the proof of Theorem 10.

Theorem 12 (The rate–content region). *The closure of the achievable rate–content pairs at distortion D is the closed convex set whose Pareto frontier is traced by $\alpha \in [0, 1]$ as follows. Let $(a, b, n, \gamma_0, \gamma_1)$, with $n > 0$ and $\gamma_0, \gamma_1 \in (0, 1)$, solve the stationarity system*

$$a = 1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1, \quad \mu_c = \frac{a\rho}{s - (1 - \alpha)\gamma_1}, \quad b = (1 - \alpha)\gamma_1\mu_c, \quad (30)$$

$$n = D - h(a, b), \quad \gamma_0 = \frac{n}{Q_0 + n}, \quad \gamma_1 = \frac{n}{Q_1 + n}. \quad (31)$$

The solution exists and is unique at every α (Proposition 14; at $\alpha = 0$ uniqueness also follows from Theorem 10), and the frontier point is

$$(R(\alpha), L(\alpha)) = \left(\frac{1}{2} \log_2 \frac{1}{\gamma_0}, \frac{1}{2} \log_2 \frac{1}{\gamma_1} \right),$$

achieved by $\hat{Y} = aY + bV + N$, $N \sim \mathcal{N}(0, n)$. Endpoints: $\alpha = 1$ recovers the classical reverse channel ($b = 0$, $a = 1 - D$, $\gamma_0 = D$, $R = \frac{1}{2} \log_2(1/D)$); $\alpha = 0$ recovers the channel and value of Theorem 10 ($\gamma_1 = 1/g^$).*

Proof. Convexity and scalarization. Both coordinates are convex in the test channel: Lemma 2 and the per- s decomposition (9) apply verbatim with the sum over s replaced by an integral against the law of S , the conditional laws $p(t | s)$ being furnished by disintegration through a regular conditional probability (S takes values in a standard Borel space). The achievable set of pairs at distortion D is therefore convex, closed, and upward closed in each coordinate, so every point of its Pareto frontier minimizes one of the weighted objectives

$$J_\alpha := \alpha I(T; \hat{Y}) + (1 - \alpha) I(T; \hat{Y} | S), \quad \alpha \in [0, 1],$$

over admissible channels, by the supporting-hyperplane argument. At the interior weights each J_α -minimizer is a frontier point; at the endpoint weights $\alpha \in \{0, 1\}$ this a priori fixes only one coordinate of the minimizer, and the value of the other coordinate is pinned by the uniqueness of the minimizing moments (Proposition 14; the argument is run in detail in Corollary 15’s proof).

The two determinant identities. Fix an admissible channel, centered without loss, with regression coordinates (a, b, n) , $n > 0$, as in the proof of Theorem 10; the Markov moment identity $\text{Cov}(N', S) = 0$ holds as there. Lemma 8 applies twice to the same three moments, once with the conditioner S and once with S deleted:

$$I(T; \hat{Y}) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_T}{\det \Sigma_{e0}} = \frac{1}{2} \log_2 \frac{Q_0 + n}{n}, \quad I(T; \hat{Y} | S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{1}{2} \log_2 \frac{Q_1 + n}{n}, \quad (32)$$

where Σ_{e0} is the error covariance of the best linear estimate of T from $\hat{Y}$ alone. The conditional identity is written out in the proof of Theorem 10. The rate identity, in full: with $\hat{c} := \mathbb{E}[T\hat{Y}] = \Sigma_T(a, b)^\top$ and $\text{Var } \hat{Y} = Q_0 + n$,

$$\Sigma_{e0} = \Sigma_T - \frac{\hat{c} \hat{c}^\top}{Q_0 + n}, \quad \det \Sigma_{e0} = \det \Sigma_T \left(1 - \frac{\hat{c}^\top \Sigma_T^{-1} \hat{c}}{Q_0 + n}\right) = \det \Sigma_T \frac{n}{Q_0 + n},$$

by the rank-one determinant identity $\det(\Sigma - \hat{c} \hat{c}^\top / v) = \det \Sigma (1 - \hat{c}^\top \Sigma^{-1} \hat{c} / v)$ and $\hat{c}^\top \Sigma_T^{-1} \hat{c} = (a, b) \Sigma_T (a, b)^\top = Q_0$.

The weighted moment program. Combining, J_α is bounded below by

$$\min_{(a,b,n)} \varphi_\alpha(a, b, n) := \frac{\alpha}{2} \log_2 \frac{Q_0 + n}{n} + \frac{1-\alpha}{2} \log_2 \frac{Q_1 + n}{n} \quad \text{s.t.} \quad h(a, b) + n \leq D, \quad n > 0. \quad (33)$$

As in Theorem 10's proof: φ_α is strictly decreasing in n at fixed $(a, b) \neq (0, 0)$, so the distortion constraint is active at the optimum; on the feasible set $Q_0 \geq Q_1 \geq q_0 > 0$, so $\varphi_\alpha \rightarrow \infty$ as $n \rightarrow 0$ and the minimum is attained at an interior stationary point of $\varphi_\alpha(a, b, D - h(a, b))$.

The gradient identities reduce the first-order system linearly. At a stationary point set $\gamma_0 = n/(Q_0 + n)$ and $\gamma_1 = n/(Q_1 + n)$, both in $(0, 1)$. For $x \in \{a, b\}$, using $\partial_x n = -\partial_x h$,

$$\alpha \frac{\partial_x(Q_0 - h)}{Q_0 + n} + (1 - \alpha) \frac{\partial_x(Q_1 - h)}{Q_1 + n} + \frac{\partial_x h}{n} = 0;$$

multiplying by $n/2$ and inserting the four gradient identities (24):

$$x = a: \quad \alpha \gamma_0 + (1 - \alpha) \gamma_1 (1 - \rho \mu_c) + (a + b\rho - 1) = 0, \quad (34)$$

$$x = b: \quad \alpha \gamma_0 \rho + (1 - \alpha) \gamma_1 (\rho - \mu_c) + (a\rho + b - \rho) = 0. \quad (35)$$

Subtract ρ times (34) from (35): the γ_0 terms cancel, the γ_1 terms give $(1 - \alpha) \gamma_1 [(\rho - \mu_c) - \rho(1 - \rho \mu_c)] = -(1 - \alpha) \gamma_1 \mu_c (1 - \rho^2)$, and the moment terms give $(a\rho + b) - \rho(a + b\rho) = b(1 - \rho^2)$, so

$$b(1 - \rho^2) = (1 - \alpha) \gamma_1 \mu_c (1 - \rho^2) \implies b = (1 - \alpha) \gamma_1 \mu_c.$$

Equation (34) then reads $a + b\rho = 1 - \alpha \gamma_0 - (1 - \alpha) \gamma_1 + (1 - \alpha) \gamma_1 \rho \mu_c = 1 - \alpha \gamma_0 - (1 - \alpha) \gamma_1 + b\rho$, so $a = 1 - \alpha \gamma_0 - (1 - \alpha) \gamma_1$. Finally $s\mu_c = a\rho + b = a\rho + (1 - \alpha) \gamma_1 \mu_c$ rearranges to $\mu_c = a\rho / (s - (1 - \alpha) \gamma_1)$, the denominator being positive since $s > 1 > (1 - \alpha) \gamma_1$. This is (30); (31) restates the activity of the constraint and the definitions of (γ_0, γ_1) . Conversely, every solution of the system with $n > 0$ is a stationary point of the reduced program, and by Proposition 14 the program is convex in moment coordinates with a unique minimizer, so the system's solution is unique and is that minimizer.

Achievability. The jointly Gaussian channel $\hat{Y} = aY + bV + N$, $N \sim \mathcal{N}(0, n)$, with the optimal (a, b, n) realizes the same moments and, by the equality clause of Lemma 8, attains both bounds of (32) simultaneously: $I(T; \hat{Y}) = \frac{1}{2} \log_2(1/\gamma_0)$ and $I(T; \hat{Y} | S) = \frac{1}{2} \log_2(1/\gamma_1)$. Hence the value of (33) equals the minimum of J_α , the frontier point is as displayed, and the trace over $\alpha \in [0, 1]$ is the entire Pareto boundary.

Endpoints. At $\alpha = 1$, (30) gives $b = 0$ and $a = 1 - \gamma_0$; then $h = (1 - a)^2 = \gamma_0^2$, $Q_0 = a^2$, and $\gamma_0(Q_0 + n) = n$ forces $\gamma_0 a^2 = n(1 - \gamma_0) = na$, so $n = \gamma_0 a$; combined with $n = D - \gamma_0^2$ this gives $\gamma_0(1 - \gamma_0) = D - \gamma_0^2$, that is, $\gamma_0 = D$, $a = 1 - D$, $n = D(1 - D)$: the classical reverse channel. At $\alpha = 0$ the γ_0 terms drop from (34)–(35), which become (25) with $g = 1/\gamma_1$; Theorem 10 identifies $\gamma_1 = 1/g^*$ and the channel (20). $\square$

Uniqueness at every weight is a convexity statement in moment coordinates. It is cleanest, and no harder, in the whitened frame that Section VII also uses; we set it up once. Since $\Sigma_T \succ 0$ and $0 \prec \Sigma_{T|S} \preceq \Sigma_T$, there is a nonsingular W with

$$W \Sigma_T W^\top = I, \quad W \Sigma_{T|S} W^\top = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p), \quad 0 \prec \Lambda \preceq I \quad (36)$$

(factor $\Sigma_T = LL^\top$, diagonalize $L^{-1}\Sigma_T L^{-\top} = O\Lambda O^\top$, and set $W = O^\top L^{-1}$). Write $Z = WT$, so $\text{Cov } Z = I$; the read becomes $Y = y_0^\top Z$ with $y_0 = W^{-\top}e_Y$ and $|y_0|^2 = e_Y^\top \Sigma_T e_Y = 1$. A channel with regression vector $(a, b)^\top$ on T has regression vector $c = W^{-\top}(a, b)^\top$ on Z , and

$$Q_0 = |c|^2, \quad Q_1 = c^\top \Lambda c, \quad h = |y_0 - c|^2, \quad \mathbb{E}(Y - \hat{Y})^2 = |y_0 - c|^2 + n.$$

Here $p = 2$; Section VII uses $p = 1 + r$.

Lemma 13 (Matrix convexity of the Schur ratio). *Let $\mathcal{Z} \subseteq \mathbb{R}^q$ be convex, $a : \mathcal{Z} \rightarrow \mathbb{R}^p$ affine, and $\sigma : \mathcal{Z} \rightarrow (0, \infty)$ concave. Then*

$$F(z) := \frac{a(z)a(z)^\top}{\sigma(z)}$$

is matrix-convex: $F(z_t) \preceq (1-t)F(z_0) + tF(z_1)$ for all $z_0, z_1 \in \mathcal{Z}$, $t \in [0, 1]$, $z_t = (1-t)z_0 + tz_1$.

Proof. For $\sigma > 0$ the Schur-complement criterion states

$$\begin{bmatrix} W & a \\ a^\top & \sigma \end{bmatrix} \succeq 0 \iff W \succeq \frac{aa^\top}{\sigma},$$

so $F(z)$ is the smallest matrix admitted by the certificate at z . Fix z_0, z_1, t and abbreviate $a_i = a(z_i)$, $\sigma_i = \sigma(z_i)$, $F_i = F(z_i)$. The two certificates

$$\begin{bmatrix} F_0 & a_0 \\ a_0^\top & \sigma_0 \end{bmatrix} \succeq 0, \quad \begin{bmatrix} F_1 & a_1 \\ a_1^\top & \sigma_1 \end{bmatrix} \succeq 0$$

average, with weights $(1-t, t)$, to

$$\begin{bmatrix} (1-t)F_0 + tF_1 & a(z_t) \\ a(z_t)^\top & (1-t)\sigma_0 + t\sigma_1 \end{bmatrix} \succeq 0,$$

using affinity of a in the off-diagonal blocks. Concavity gives $\sigma(z_t) \geq (1-t)\sigma_0 + t\sigma_1$, and adding the positive semidefinite matrix $\text{diag}(0, \sigma(z_t) - (1-t)\sigma_0 - t\sigma_1)$ raises the corner:

$$\begin{bmatrix} (1-t)F_0 + tF_1 & a(z_t) \\ a(z_t)^\top & \sigma(z_t) \end{bmatrix} \succeq 0.$$

By the criterion again, $(1-t)F_0 + tF_1 \succeq a(z_t)a(z_t)^\top/\sigma(z_t) = F(z_t)$. $\square$

Proposition 14 (Uniqueness at every weight). *In the whitened frame (36), with $0 \prec \Lambda \preceq I$, $|y_0| = 1$, $D \in (0, 1)$, and any $\alpha \in [0, 1]$, the weighted program (33) (with $Q_0 = |c|^2$, $Q_1 = c^\top \Lambda c$, $h = |y_0 - c|^2$, minimized over (c, n)) has a unique minimizer, and every stationary point with $n > 0$ is that minimizer. Since the change of variables $c = W^{-\top}(a, b)^\top$ is a linear bijection, the same holds for the program in the coordinates of Theorem 12.*

Proof. Work in the moment coordinates $z = (c, v)$, $v = \text{Var } \hat{Y} = |c|^2 + n$, on the domain

$$\mathcal{D} = \{(c, v) : v > |c|^2, 1 - 2y_0^\top c + v \leq D\},$$

which is convex: $v > |c|^2$ is the strict epigraph of a convex function, and the distortion constraint $h + n = 1 - 2y_0^\top c + v \leq D$ is linear in z . On $\mathcal{D}$ define $\sigma := v - c^\top(I - \Lambda)c = Q_1 + n > 0$ and

$$B_0 = \frac{1}{2} \log_2 \frac{Q_0 + n}{n} = -\frac{1}{2} \log_2 \det \left(I - \frac{cc^\top}{v} \right),$$

$$B_1 = \frac{1}{2} \log_2 \frac{Q_1 + n}{n} = \frac{1}{2} \log_2 \det \Lambda - \frac{1}{2} \log_2 \det \left(\Lambda - \frac{(\Lambda c)(\Lambda c)^\top}{\sigma} \right);$$

the first equality holds because $\det(I - cc^\top/v) = 1 - |c|^2/v = n/v$, the second because $\det(\Lambda - (\Lambda c)(\Lambda c)^\top/\sigma) = \det \Lambda (1 - c^\top \Lambda c/\sigma) = \det \Lambda \cdot n/\sigma$.

Convexity. Apply Lemma 13 twice. For B_0 : $z \mapsto c$ is affine and $z \mapsto v$ is affine, hence concave and positive on $\mathcal{D}$, so $W_0(z) = cc^\top/v$ is matrix-convex, $I - W_0$ is matrix-concave, and $I - W_0 \succ 0$ on $\mathcal{D}$ (Schur criterion with $v - |c|^2 = n > 0$). For B_1 : $z \mapsto \Lambda c$ is affine and $z \mapsto \sigma$ is concave (affine minus a convex quadratic, $I - \Lambda \succeq 0$) and positive, so $W_1(z) = (\Lambda c)(\Lambda c)^\top/\sigma$ is matrix-convex, $\Lambda - W_1$ is matrix-concave, and $\Lambda - W_1 \succ 0$

since $c^\top \Lambda c = Q_1 < \sigma$. The map $\Sigma \mapsto -\log_2 \det \Sigma$ on positive definite matrices is convex and antitone in the semidefinite order, and composing a convex antitone function with a matrix-concave map preserves convexity: for matrix-concave $F \succ 0$,

$$-\log \det F(z_t) \leq -\log \det((1-t)F(z_0) + tF(z_1)) \leq (1-t)(-\log \det F(z_0)) + t(-\log \det F(z_1)).$$

Hence B_0 , B_1 , and every $B_\alpha := \alpha B_0 + (1-\alpha)B_1$ are convex on $\mathcal{D}$.

Reduction to the active slice. On $\mathcal{D}$, feasibility forces $c \neq 0$: from $v > |c|^2 \geq 0$ and $1 - 2y_0^\top c + v \leq D$,

$$2y_0^\top c \geq 1 - D + v > 1 - D > 0. \quad (37)$$

At fixed $c \neq 0$ both B_0 and B_1 strictly decrease in v ($\partial_v[\ln v - \ln n] = v^{-1} - n^{-1} < 0$ and $\partial_v[\ln \sigma - \ln n] = \sigma^{-1} - n^{-1} < 0$, since $Q_0 \geq Q_1 > 0$), so every minimizer of B_α lies on the active slice $v = D - 1 + 2y_0^\top c$, an affine section of $\mathcal{D}$. On the slice B_α is convex, so its stationary points are exactly its global minimizers and the minimizer set is convex; coercivity as in Theorem 12's proof gives attainment.

Strictness. Suppose $z_0 \neq z_1$ are both minimizers on the slice, and let z_t run over the segment, along which v and c are affine in t . Then B_α is affine on the segment, and since B_0 , B_1 are convex and their weighted sum is affine, each B_i carrying nonzero weight is affine on the segment. We use twice the following fact: for an affine positive definite matrix segment $\Sigma_t = (1-t)\Sigma_0 + t\Sigma_1$, the function $t \mapsto -\log \det \Sigma_t$ has second derivative $\text{tr}[(\Sigma_t^{-1} \dot{\Sigma})^2] \geq 0$, which vanishes only if $\dot{\Sigma} = 0$, because $\Sigma_t^{-1/2} \dot{\Sigma} \Sigma_t^{-1/2}$ is symmetric and a symmetric matrix whose square has zero trace is zero. Moreover, if a matrix-concave F has $-\log \det F$ affine on the segment, then squeezing $-\log \det F(z_t) \leq -\log \det((1-t)F(z_0) + tF(z_1)) \leq$ the affine interpolant forces both inequalities to be equalities; the second forces $F(z_0) = F(z_1) =: \bar{F}$ by the fact just stated, and the first then forces $F(z_t) \succeq \bar{F}$ with $\det F(z_t) = \det \bar{F}$, hence $F(z_t) = \bar{F}$ for all t (if $A \succeq B \succ 0$ and $\det A = \det B$ then $A = B$).

Case $\alpha > 0$: B_0 is affine, so $W_0(z_t) = c(t)c(t)^\top/v(t)$ is constant, a rank-one matrix $\bar{W}_0 \neq 0$ by (37). Then $c(t) = \theta(t)e$ for a fixed unit vector e , with θ affine (a coordinate of the affine c), and $\theta(t)^2 = v(t)(e^\top \bar{W}_0 e)$ with v affine. An affine function whose square is affine is constant, so θ , hence c , hence $v = D - 1 + 2y_0^\top c$, is constant: $z_0 = z_1$, a contradiction.

Case $\alpha < 1$: B_1 is affine, so $W_1(z_t) = (\Lambda c)(\Lambda c)^\top/\sigma$ is constant and nonzero. Then $\Lambda c(t) = \theta(t)e$ with θ affine, and $\theta(t)^2 = \sigma(t)(e^\top \bar{W}_1 e)$. Along the segment $\sigma(t) = v(t) - c(t)^\top(I - \Lambda)c(t)$ is a concave quadratic in t , while θ^2 is a convex quadratic; equality up to the positive constant $e^\top \bar{W}_1 e$ forces both to be affine, so θ is an affine function with affine square, hence constant. Then Λc is constant, c is constant (Λ nonsingular), and $z_0 = z_1$ as before, a contradiction.

Every $\alpha \in [0, 1]$ falls in at least one case, so the minimizer is unique. $\square$

Corollary 15 (Misalignment opens a strict tradeoff). *For $\rho^2 \in (0, 1)$ and $\tau^2 > 0$ the frontier is nondegenerate: any channel attaining $L(D)$ has rate strictly above $R_{\min} = \frac{1}{2} \log_2(1/D)$, and any channel attaining $R_{\min}$ has conditional content strictly above $L(D)$. In particular*

$$L(1) = \frac{1}{2} \log_2 \frac{(1-D)(1-\rho^2/s) + D}{D} > L(D),$$

the content of the classical reverse channel, and $R(0) = \frac{1}{2} \log_2[(Q_0+n)/n]$ evaluated at the channel of Theorem 10 strictly exceeds $R_{\min}$. (Numerically, at $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$: $R(0) - R_{\min} = 0.0400$ and $L(1) - L(D) = 0.0349$ bits.) The collapse degeneracies are exactly the alignment boundaries: at $\rho = 0$ the classical reverse channel attains both minima and the frontier is a single corner, and the frontier collapses in the limits $\rho^2 \rightarrow 1$ (the scalar corner of [16]) and $\tau^2 \rightarrow \infty$ (where $L \rightarrow R$ coordinatewise).

Proof. By Theorem 10, the moment program for the conditional coordinate alone has the unique minimizer (a^*, b^*, n^*) of (27), and $b^* = (g^* - 1)\rho/(g^*k) \neq 0$ since $\rho \neq 0$, $g^* > 1$, $k > 0$. By the $\alpha = 1$ endpoint computation in Theorem 12's proof (uniqueness by Proposition 14), the moment program for the rate coordinate alone has the unique minimizer $(1-D, 0, D(1-D))$, with $b = 0$. The two minimizers are distinct.

Let a channel attain $L(D)$. Writing B_0, B_1 for the two bounds of (32) as functions of the channel's moments, $L(D) = I(T; \hat{Y} | S) \geq B_1(\text{its moments}) \geq \min B_1 = L(D)$ forces equality throughout, so its moments minimize B_1 and equal (a^*, b^*, n^*) . Its rate then satisfies $R \geq B_0(a^*, b^*, n^*) > \min B_0 = R_{\min}$: strict, because B_0 's minimizer is unique and differs from (a^*, b^*, n^*) . Symmetrically, any rate-optimal channel carries the moments

$(1 - D, 0, D(1 - D))$ and its content is at least B_1 there, which strictly exceeds $L(D)$; evaluating, $b = 0$ gives $Q_1 = (1 - D)^2(1 - \rho^2/s)$ and $n = D(1 - D)$, so $(Q_1 + n)/n = [(1 - D)(1 - \rho^2/s) + D]/D$, the displayed $L(1)$.

At $\rho = 0$, $b^* = 0$ and $g^* = 1/D$ (Corollary 11(ii)) give $(a^*, n^*) = (1 - D, D(1 - D))$: the two minimizers coincide and the frontier is the corner $R_{\min} = L(D) = \frac{1}{2} \log_2(1/D)$. As $\rho^2 \rightarrow 1$ the whole frontier converges to the scalar corner of [16], and as $\tau^2 \rightarrow \infty$, $\Sigma_{T|S} \rightarrow \Sigma_T$ makes the two coordinates, and so the two endpoints, coincide in the limit. $\square$

Corollary 16 (L is not a functional of the (Y, S) law). *Two instances with the same (Y, S) joint law can have different minimal conditional content: the instances $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$ and $(\frac{3}{4}, \frac{1}{2})$ have identical (Y, S) margins ($\rho_{YS}^2 = \frac{1}{2}$) yet $L = 1.1610$ versus 1.2105 bits at $D = 0.1$, and $L = 0.3685$ versus 0.5228 bits at $D = 0.3$, in both cases strictly below the Steinberg value of the margin (1.2297 and 0.5577 bits respectively). In particular $L(D)$ is not Steinberg's function composed with any modified pair, nor Gray's, nor any functional of the (Y, S) joint law: the encoder's access to the clean context V is load-bearing.*

Proof. The two instances share the (Y, S) joint law: in both, Y is standard and $\rho_{YS}^2 = \text{Cov}(Y, S)^2 / \text{Var } S = \rho^2/s$ equals $\frac{1}{2}$, namely $(\frac{1}{2})/1$ and $(\frac{3}{4})/(\frac{3}{2})$, and rescaling S changes none of the quantities considered.

First instance, $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$: the context is clean. This instance sits at the boundary $\tau^2 = 0$ excluded by the standing assumption $\tau^2 > 0$ of Section III, so its value is supplied not by Theorem 10 but directly by Proposition 7(ii), which is proved at $\tau^2 = 0$ (and which the $\tau^2 \rightarrow 0$ anchor of Corollary 11 matches in the limit): $L = \frac{1}{2} \log_2(1/(2D))$: $L = \frac{1}{2} \log_2 5 = 1.1610$ bits at $D = 0.1$, and $L = \frac{1}{2} \log_2(5/3) = 0.3685$ bits at $D = 0.3$.

Second instance, $(\rho^2, \tau^2) = (\frac{3}{4}, \frac{1}{2})$: $s = \frac{3}{2}$, and (19) becomes, at $D = 0.1$,

$$P(g) = \frac{3}{20}g^2 - \frac{17}{20}g + \frac{1}{4}, \quad g^* = \frac{17 + \sqrt{229}}{6} = 5.3555, \quad L = \frac{1}{2} \log_2 g^* = 1.2105 \text{ bits},$$

and, at $D = 0.3$,

$$P(g) = \frac{9}{20}g^2 - \frac{21}{20}g + \frac{1}{4}, \quad g^* = \frac{21 + \sqrt{261}}{18} = 2.0642, \quad L = \frac{1}{2} \log_2 g^* = 0.5228 \text{ bits}.$$

The Steinberg value of the margin: an encoder observing Y alone pays the marginalized-class optimum of Section IV, the scalar-corner value $\frac{1}{2} \log_2[(1 - \rho_{YS}^2 + \rho_{YS}^2 D)/D]$ at $\rho_{YS}^2 = \frac{1}{2}$, that is, $\frac{1}{2} \log_2[(1 + D)/(2D)]$: $\frac{1}{2} \log_2 5.5 = 1.2297$ bits at $D = 0.1$, and $\frac{1}{2} \log_2(13/6) = 0.5577$ bits at $D = 0.3$. Both instances lie strictly below it ($1.1610, 1.2105 < 1.2297$ and $0.3685, 0.5228 < 0.5577$), as Proposition 7 requires.

Since the two instances have the same (Y, S) law and different L , no functional of the (Y, S) joint law, in particular neither Steinberg's function nor Gray's under any reparameterization of the pair, can equal L . $\square$

VI. WHEN THE FLOOR IS ATTAINED

The conditional content of any admissible description is floored by a determinant ratio, and it is natural to ask when the floor is exact. The question is sharpest for a joint description serving two reads, and this section settles it there; the single-read answer appears as a corollary inside the proof. In this section only, $Y = (Y_A, Y_B)$ denotes a pair of normalized reads of the source, jointly Gaussian with the normalized context V , with an arbitrary correlation pattern; the third party again holds $S = V + U$, $U \sim \mathcal{N}(0, \tau^2)$, with X , U , and all channel randomness mutually independent. A joint channel produces $\hat{Y} = (\hat{Y}_A, \hat{Y}_B)$ with $\hat{Y} - X - S$ and must meet $\mathbb{E}(Y_A - \hat{Y}_A)^2 \leq D_A$ and $\mathbb{E}(Y_B - \hat{Y}_B)^2 \leq D_B$; write $L_{\min}(D_A, D_B) = \min I(X; \hat{Y} | S)$.

Write Σ_Y for the read-pair covariance, $\beta = \text{Cov}(Y, V) \in \mathbb{R}^2$, $\Sigma_{Y|V} = \Sigma_Y - \beta\beta^\top$, $\Sigma_{Y|S} = \Sigma_Y - \beta\beta^\top/(1 + \tau^2)$, and $\Delta = \text{diag}(D_A, D_B) \succ 0$. Every admissible channel obeys the *work floor*

$$L \geq I(Y; \hat{Y} | S) = h(Y | S) - h(Y | \hat{Y}, S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{Y|S}}{\det \Delta}, \quad (38)$$

since $h(Y | S) = \frac{1}{2} \log_2((2\pi e)^2 \det \Sigma_{Y|S})$ and $h(Y | \hat{Y}, S) \leq \sum_{i \in \{A, B\}} h(Y_i - \hat{Y}_i) \leq \frac{1}{2} \log_2((2\pi e)^2 D_A D_B)$.

Three facts from the machinery already built carry over. First, pair sufficiency extends verbatim: the resampling proof of Theorem 6 applies with $T = (Y_A, Y_B, V)$, since $S = V + U$ with U jointly independent of $(X, \text{all channel randomness})$ and the distortions are functionals of the pairs $(Y_i, \hat{Y}_i)$; so the channel may be restricted to depend on X only through T , and then $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. Second, the same mutual independence

forces the Markov moment identity $\text{Cov}(S, \hat{\mathbf{Y}}) = \text{Cov}(V, \hat{\mathbf{Y}})$. Third, jointly Gaussian channels $\hat{\mathbf{Y}} = A\mathbf{T} + N$, $N \sim \mathcal{N}(0, \Sigma_N)$, exhaust the problem, by Lemma 8: for any admissible T -channel with $C = \text{Cov}(T, \hat{\mathbf{Y}})$ and $\Sigma_{\hat{\mathbf{Y}}} = \text{Cov}(\hat{\mathbf{Y}})$, set $A := C^T \Sigma_T^{-1}$ and $\Sigma_N := \Sigma_{\hat{\mathbf{Y}}} - A \Sigma_T A^T \succeq 0$ (assume $\Sigma_T \succ 0$; degenerate instances reduce by dropping dependent context components; singular Σ_N is dispatched by Remark 9). The residual $\hat{\mathbf{Y}} - A\mathbf{T}$ is uncorrelated with T , hence with V , and with U by mutual independence, hence with S , so the lemma's hypothesis holds and

$$L \geq \frac{1}{2} \log_2 \frac{\det(A \Sigma_{T|S} A^T + \Sigma_N)}{\det \Sigma_N}, \quad \Sigma_{T|S} = \Sigma_T - \frac{\sigma_{TS} \sigma_{TS}^T}{1 + \tau^2}, \quad \sigma_{TS} = \Sigma_{T^c V}, \quad (39)$$

with equality for the jointly Gaussian channel realizing the same moments, and the distortion constraints $(e_i - A_{i\cdot})^T \Sigma_T (e_i - A_{i\cdot}) + (\Sigma_N)_{ii} \leq D_i$ depend on the channel only through those moments. Consequently $L_{\min}$ is the minimum of (39) over a compact feasible set of pairs (A, Σ_N) and is attained.

A plausible conjecture states the attainment condition for (38) through the third party's conditional covariance $\Sigma_{Y|S}$; the truth involves the encoder-accessible $\Sigma_{Y|V}$, the same S -versus- V substitution as in Proposition 7.

Theorem 17 (Work-floor exactness). *The work floor $\frac{1}{2} \log_2(\det \Sigma_{Y|S} / \det \Delta)$ of (38) is attained iff*

$$(a) \tau^2 = 0 \text{ and } \Delta \preceq \Sigma_{Y|V}, \quad \text{or} \quad (b) \beta = 0 \text{ and } \Delta \preceq \Sigma_Y.$$

In every other case the floor is strict, both on the misaligned branch ($\tau^2 > 0$ and $\beta \neq 0$, at every positive (D_A, D_B)) and outside the semidefinite conditions of (a) and (b). On the misaligned branch with $\Delta \prec \Sigma_{Y|V}$, the deficit obeys the sandwich

$$0 < L_{\min} - \text{floor} \leq \frac{1}{2} \log_2 \frac{\det(I - \Sigma_{Y|S}^{-1} \Delta)}{\det(I - \Sigma_{Y|V}^{-1} \Delta)} = \frac{1}{2} \text{tr}[(\Sigma_{Y|V}^{-1} - \Sigma_{Y|S}^{-1}) \Delta] \log_2 e + O(\|\Delta\|^2) \rightarrow 0 \quad (\Delta \rightarrow 0).$$

Proof. Necessity. Suppose the floor is attained. Equality in the chain (38) forces, at the moment level (max-entropy tightness makes the joint WLOG Gaussian): $E_S := \mathbb{E}[\text{Cov}(Y | \hat{\mathbf{Y}}, S)] = \Delta$ with per-consumer equality, so $Z := Y - \hat{\mathbf{Y}}$ has $\text{Cov} Z = \Delta$, $\text{Cov}(Z, \hat{\mathbf{Y}}) = 0$, $\text{Cov}(Z, S) = 0$; the latter with the Markov moment identity gives $\text{Cov}(Z, V) = 0$, hence $\text{Cov}(\hat{\mathbf{Y}}, V) = \beta$. Since $E_S \preceq \Sigma_{Y|S}$ always, $\Delta \preceq \Sigma_{Y|S}$ is necessary; at $\tau^2 = 0$ this is $\Delta \preceq \Sigma_{Y|V}$. Tightness of $I(T; \hat{\mathbf{Y}} | S) = I(Y; \hat{\mathbf{Y}} | S)$ additionally requires $\text{Cov}(V, \hat{\mathbf{Y}} | Y, S) = 0$; with the pinned moments this evaluates to $[\Delta \ 0] M^{-1} [\beta; \text{Cov}(S, V)] = 0$, $M = \text{Cov}(Y, S)$, which (for $\Delta \succ 0$) holds iff $\beta = 0$ or $\text{Cov}(S, V) = \text{Var } S$, i.e. $\tau^2 = 0$.

Sufficiency (a). At $\tau^2 = 0$ take $Z \sim \mathcal{N}(0, \Delta)$ jointly Gaussian with $\text{Cov}(Z, Y) = \Delta$, $\text{Cov}(Z, V) = 0$; such a Z exists iff $\Delta - \Delta \Sigma_{Y|V}^{-1} \Delta \succeq 0 \Leftrightarrow \Delta \preceq \Sigma_{Y|V}$ (Schur complement). Set $\hat{\mathbf{Y}} = Y - Z$: then $\text{Cov}(\hat{\mathbf{Y}} | S) = \Sigma_{Y|S} - \Delta$ and $\text{Cov}(\hat{\mathbf{Y}} | T, S) = \Delta - \Delta \Sigma_{Y|V}^{-1} \Delta$, and at $\tau^2 = 0$ (where $\Sigma_{Y|S} = \Sigma_{Y|V} =: \Sigma_c$)

$$L = \frac{1}{2} \log_2 \frac{\det(\Sigma_c - \Delta)}{\det \Delta \det(I - \Sigma_c^{-1} \Delta)} = \frac{1}{2} \log_2 \frac{\det \Sigma_c}{\det \Delta} = \text{floor}.$$

Sufficiency (b). At $\beta = 0$, $\Sigma_{Y|S} = \Sigma_Y$ and a channel depending on X only through Y has $\hat{\mathbf{Y}} \perp S$, so $L = I(Y; \hat{\mathbf{Y}})$, attained at the floor iff $\Delta \preceq \Sigma_Y$ (the exactness condition of the bivariate rate-distortion function under individual distortion constraints [10]); necessity of $\Delta \preceq \Sigma_Y$ is the $E_S \preceq \Sigma_{Y|S} = \Sigma_Y$ bound.

Sandwich. The same Z -construction at $\tau^2 > 0$ is admissible whenever $\Delta \prec \Sigma_{Y|V}$ and yields $L \leq \frac{1}{2} \log_2[\det(\Sigma_{Y|S} - \Delta) / \det(\Delta - \Delta \Sigma_{Y|V}^{-1} \Delta)] = \text{floor} + \frac{1}{2} \log_2[\det(I - \Sigma_{Y|S}^{-1} \Delta) / \det(I - \Sigma_{Y|V}^{-1} \Delta)]$; the ratio is ≥ 1 because $\Sigma_{Y|S} \succeq \Sigma_{Y|V}$ (congruence by $\Delta^{1/2}$ keeps both factors in the PSD order where $\det$ is monotone), and $\rightarrow 1$ as $\Delta \rightarrow 0$. Strictness (> 0) is the necessity direction, using that $L_{\min}$ is attained (compact feasible (A, Σ_N) set, lower semicontinuity); at the non-strict boundary of $\Delta \preceq \Sigma_{Y|V}$ in case (a), where the displayed ratio is $0/0$, attainment follows by $\Delta_t \uparrow \Delta$ feasibility and floor continuity.

The single-read corollary. For one consumer, on the active-floor regime $D \leq \text{Var}(Y | V)$, the floor $\frac{1}{2} \log_2(\text{Var}(Y | S)/D)$ is attained iff $\rho\tau = 0$: substituting $g_f = (s - \rho^2)/(Ds)$ into the quadratic $P(g)$ of Theorem 10, equation (19), whose larger root g^* gives $L(D) = \frac{1}{2} \log_2 g^*$, yields $P(g_f) = -\rho^2 \tau^2 / s$, which is < 0 (hence $g^* > g_f$ strictly) exactly when $\rho\tau \neq 0$. (The regime qualifier is the single-read shadow of condition (a): for $\tau = 0$, $D > 1 - \rho^2$ the clamped floor and $L = 0$ meet trivially while $g_f < 1$ becomes the quadratic's smaller root.) $\square$

VII. VECTOR CONTEXT

The frontier survives a vector conditioner. Let $S = \Gamma^\top X + U \in \mathbb{R}^r$, $U \sim \mathcal{N}(0, \Sigma_U)$, $\Sigma_U \succ 0$, with X , U , and all channel randomness mutually independent. Write $V := \Gamma^\top X \in \mathbb{R}^r$ and $T := (Y, V) \in \mathbb{R}^{1+r}$, and assume $\Sigma_T \succ 0$; degenerate instances reduce by dropping linearly dependent context components. Reproductions are scalar without loss of optimality by the data-processing reduction of Section III, which is generic in S .

Theorem 18 (Vector context). *Let $0 < D < 1$. (i) Sufficiency: it is without loss of optimality to let the channel depend on X only through T , and then $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. (ii) Whitened form: with W , $Z = WT$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_{1+r})$, $0 \prec \Lambda \preceq I$, and the unit vector y_0 of (36) at $p = 1 + r$, every admissible channel with regression vector $c = \mathbb{E}[Z\hat{Y}]$ and residual variance $n > 0$ satisfies*

$$R \geq \frac{1}{2} \log_2 \frac{|c|^2 + n}{n}, \quad L \geq \frac{1}{2} \log_2 \frac{c^\top \Lambda c + n}{n}, \quad \mathbb{E}(Y - \hat{Y})^2 = |y_0 - c|^2 + n, \quad (40)$$

with both bounds attained simultaneously by the jointly Gaussian channel $\hat{Y} = c^\top Z + N$, $N \sim \mathcal{N}(0, n)$. (iii) The frontier, traced by $\alpha \in [0, 1]$: with $\gamma_0 = n/(|c|^2 + n)$ and $\gamma_1 = n/(c^\top \Lambda c + n)$, the Pareto-optimal channel is the unique solution with $n > 0$ of the generalized two-water-level system

$$c = (1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1) [(1 - (1 - \alpha)\gamma_1)I + (1 - \alpha)\gamma_1\Lambda]^{-1} y_0, \quad n = D - |y_0 - c|^2, \quad (41)$$

per-mode explicit given the two scalars (γ_0, γ_1) , and the frontier point is $(\frac{1}{2} \log_2(1/\gamma_0), \frac{1}{2} \log_2(1/\gamma_1))$. For $r = 1$ the program coincides with that of Theorem 12 under the linear bijection $c = W^{-\top}(a, b)^\top$, so the two systems trace the same frontier. The $\alpha = 0$ endpoint gives the vector-context function $L(D)$ as a one-dimensional root problem in γ_1 .

Proof. (i) In the resampling proof of Theorem 6, the only property of S used is that its conditional law given $(X, \hat{Y})$, and given $(X, \hat{Y}')$, is the fixed kernel $p(s | t, \hat{y}) = p_U(s - v)$, depending on x only through t . This holds verbatim for the vector $S = V + U$ with $U \sim \mathcal{N}(0, \Sigma_U)$ jointly independent of X and the channel randomness. Steps (a) through (d) of that proof are unchanged.

(ii) *Whitening.* First, $\Sigma_{T|S} = \Sigma_T - \Sigma_{TS}\Sigma_S^{-1}\Sigma_{ST} \succ 0$: positivity of $\text{Cov}((T, S))$ holds because $\Sigma_T \succ 0$ and $\Sigma_{S|T} = \text{Cov}(\Gamma^\top X | T) + \Sigma_U \succeq \Sigma_U \succ 0$, and the Schur complement of a positive definite matrix is positive definite. Also $\Sigma_{T|S} \preceq \Sigma_T$. The congruence (36) therefore exists at $p = 1 + r$ with $0 \prec \Lambda \preceq I$, and $\text{Cov} Z = I$ makes the read a unit vector: $Y = y_0^\top Z$, $|y_0|^2 = e_Y^\top \Sigma_T e_Y = 1$.

Moments and the two bounds. Fix an admissible channel, centered without loss, with $c = \mathbb{E}[Z\hat{Y}]$ and $n = \text{Var} \hat{Y} - |c|^2$; the case $n = 0$ is dispatched by Remark 9. The residual $N' = \hat{Y} - c^\top Z$ is uncorrelated with Z , hence with T and with its subvector V ; and it is uncorrelated with U by mutual independence, hence with $S = V + U$: componentwise, $\text{Cov}(N', S) = 0$. Lemma 8 applies with $A = c^\top W$: $A\Sigma_T A^\top = c^\top W\Sigma_T W^\top c = |c|^2$ and $A\Sigma_{T|S} A^\top = c^\top \Lambda c$, giving the two bounds of (40). The distortion evaluation is exact because Z is white: $\mathbb{E}(Y - \hat{Y})^2 = 1 - 2y_0^\top c + (|c|^2 + n) = |y_0 - c|^2 + n$. Equality for the Gaussian channel is Lemma 8's equality clause.

(iii) *The weighted program.* As in Theorem 12: both coordinates are convex in the channel, so the frontier is the trace of the weighted minimizations, and the weighted objective is bounded below by the moment program

$$\min_{c, n} \frac{\alpha}{2} \log_2 \frac{Q_0 + n}{n} + \frac{1 - \alpha}{2} \log_2 \frac{Q_1 + n}{n}, \quad Q_0 = |c|^2, \quad Q_1 = c^\top \Lambda c, \quad |y_0 - c|^2 + n \leq D, \quad n > 0,$$

whose value the Gaussian channel of (ii) attains. The objective is strictly decreasing in n at fixed $c \neq 0$, so the distortion constraint is active at the optimum. Feasibility forces $2y_0^\top c \geq 1 - D + v > 1 - D > 0$ as in (37), so $c \neq 0$ and $Q_1 \geq \lambda_{\min}|c|^2$ is bounded away from zero on the feasible set, giving coercivity as $n \rightarrow 0$ and attainment at an interior stationary point of the reduced objective in c , with $n = D - |y_0 - c|^2$.

The first-order system. At such a point, with $h = |y_0 - c|^2$, $\nabla h = -2(y_0 - c)$, $\nabla n = 2(y_0 - c)$, $\nabla Q_0 = 2c$, and $\nabla Q_1 = 2\Lambda c$, stationarity of $\alpha \ln(Q_0 + n) + (1 - \alpha) \ln(Q_1 + n) - \ln n$ reads, after multiplying by $n/2$,

$$\alpha\gamma_0(c + (y_0 - c)) + (1 - \alpha)\gamma_1(\Lambda c + (y_0 - c)) - (y_0 - c) = 0.$$

The first parenthesis collapses, $c + (y_0 - c) = y_0$; collecting the c terms,

$$[(1 - (1 - \alpha)\gamma_1)I + (1 - \alpha)\gamma_1\Lambda]c = (1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1)y_0,$$

which is (41). The bracket is invertible: each diagonal entry $(1 - (1 - \alpha)\gamma_1) + (1 - \alpha)\gamma_1\lambda_j$ is a convex combination of 1 and $\lambda_j > 0$, since $(1 - \alpha)\gamma_1 \in [0, 1]$. Uniqueness of the minimizer, and hence of the system's solution with $n > 0$, at every $\alpha \in [0, 1]$ is Proposition 14 at $p = 1 + r$; its proof is dimension-free. Achievability of the frontier point is (ii).

For $r = 1$ the bijection $c = W^{-\top}(a, b)^\top$ carries (Q_0, Q_1, h) and the constraint to their Section IV forms, so the two weighted programs are identical with the same unique minimizer, and the two stationarity systems are equivalent parameterizations of it. At $\alpha = 0$, (41) reads $c = (1 - \gamma_1)[(1 - \gamma_1)I + \gamma_1\Lambda]^{-1}y_0$; substituting into $n = D - |y_0 - c|^2$ and $\gamma_1 = n/(c^\top \Lambda c + n)$ leaves one scalar equation in $\gamma_1 \in (0, 1)$, the vector-context generalization of the quadratic (19). $\square$

VIII. THE BINARY SOLUTION

The binary case closes completely and, unlike its Gaussian counterpart, with an elementary self-contained proof.

Let (Y, V) be a doubly symmetric binary pair (fair marginals, $\Pr[Y \neq V] = p \in (0, \frac{1}{2})$), context V , side information $S = V \oplus W$ with $W \sim \text{Bern}(q)$, $q \in [0, \frac{1}{2}]$, independent. The record is the reproduction $\hat{Y} \in \{0, 1\}$ itself (richer records only cost more: for any stored Z from which the consumer computes $\hat{Y} = f(Z)$, $I(Y, V; Z | S) \geq I(Y, V; \hat{Y} | S)$ by the data-processing inequality), Hamming distortion $\Pr[\hat{Y} \neq Y] \leq D$. Write $x * q = x(1 - q) + (1 - x)q$ and $\ell(x) = \ln \frac{1-x}{x}$.

Theorem 19 (The binary conditional common-reconstruction function). *For $D \in (0, \frac{1}{2})$, the conditional coordinate $L(D) = \min I(Y, V; \hat{Y} | S)$ over channels $P(\hat{Y} | Y, V)$ (Markov $\hat{Y} - (Y, V) - S$) with $\Pr[\hat{Y} \neq Y] \leq D$ is attained in the two-parameter symmetric family $\hat{Y} = Y \oplus E$, $\Pr[E = 1 | Y \oplus V = j] = d_j$, and equals*

$$L(D) = h_2(u) - (1 - p)h_2(d_0) - ph_2(d_1), \quad u = a * q, \quad a = (1 - p)d_0 + p(1 - d_1),$$

where, for $q \in (0, \frac{1}{2}]$ with any $D \in (0, \frac{1}{2})$, and for $q = 0$ with $D \leq p$, the pair (d_0, d_1) solves the active constraint $(1 - p)d_0 + pd_1 = D$ together with the tilt equation

$$\ell(d_0) - \ell(d_1) = 2(1 - 2q)\ell(u).$$

(In the excluded sliver $q = 0$, $D > p$, the constraint is inactive and $L(D) = 0$, attained non-uniquely by records measurable in V alone; the tilt system still has a root there, the degenerate point $d_1 = 1 - d_0$, which evaluates to the correct value 0. For $q > 0$ the constraint is active on all of $(0, \frac{1}{2})$: L is convex, nonincreasing, positive by the last claim below, and 0 at $D = \frac{1}{2}$, hence strictly decreasing.) On the same channel the rate coordinate is $R = 1 - (1 - p)h_2(d_0) - ph_2(d_1)$ and the discount is exactly $R - L = 1 - h_2(u) = I(\hat{Y}; S)$ (the chain-rule identity $I(Y, V; \hat{Y}) - I(Y, V; \hat{Y} | S) = I(\hat{Y}; S)$, valid under the Markov chain, in closed form: $a = \Pr[\hat{Y} \neq V]$, $u = \Pr[\hat{Y} \neq S]$). Anchors: at $q = 0$ the system gives $a = (p - D)/(1 - 2D)$ and $L = h_2(p) - h_2(D)$ (Gray's conditional rate-distortion function [3], for $D \leq p$); at $q = \frac{1}{2}$ it gives $d_0 = d_1 = D$ and $L = 1 - h_2(D)$ (the marginal rate-distortion function; the context is useless exactly when the side information is). For $q \in (0, \frac{1}{2}]$ and $D < \frac{1}{2}$, $L(D) > 0$.

Proof. Reduction and convexity. Binary $\hat{Y}$ is definitional (DPI above). With $P(Y, V | S = s)$ fixed and the channel shared across s (Markov), $I(Y, V; \hat{Y} | S) = \sum_s P(s) I_s$ is a nonnegative combination of mutual informations each convex in the channel, hence convex in the channel; the distortion is linear.

Symmetrization. The global bit flip $\sigma : (y, v, s, \hat{y}) \mapsto (1 - y, 1 - v, 1 - s, 1 - \hat{y})$ preserves the joint law of (Y, V, S) and the distortion, and $L(C^\sigma) = L(C)$; averaging C with C^σ preserves distortion and, by convexity, does not increase L . So an optimum exists among σ -symmetric channels, and these are exactly the family $\{(d_0, d_1)\}$: σ -symmetry forces $\Pr[\hat{Y} \neq Y | y, v]$ to depend only on $y \oplus v$.

Closed form. On the family, $H(\hat{Y} | Y, V) = (1 - p)h_2(d_0) + ph_2(d_1)$, and $\hat{Y} \oplus S = J \oplus E \oplus W$ ($J = Y \oplus V$) with $\Pr[J \oplus E = 1] = a$. The step $H(\hat{Y} | S) = h_2(a * q)$ needs $\hat{Y} \oplus S \perp S$, which holds because the joint law of $(\hat{Y}, S)$ is σ -invariant (σ -invariant source, σ -symmetric channel), forcing equal conditional flip probabilities given either value of S ; the same symmetry gives $H(\hat{Y}) = 1$, hence R .

Tilt equation. Substituting $d_1 = (D - (1 - p)d_0)/p$ makes $a = 2(1 - p)d_0 + p - D$ affine in d_0 ; differentiating L and using $h'_2(x) = \ell(x)$ (nats) gives $(1 - p)[2(1 - 2q)\ell(u) - \ell(d_0) + \ell(d_1)] = 0$. Since L restricted to the affine family slice is convex (restriction of a convex function), interior stationarity is sufficient, not merely necessary.

Anchors and positivity. $q = \frac{1}{2}$: $u = \frac{1}{2}$, the tilt equation forces $d_0 = d_1 = D$. $q = 0$: substitution of Gray's backward channel; equality of values is elementary algebra. Positivity, in two branches over the family (which suffices by attainment): if $a \neq \frac{1}{2}$, then $h_2(u) > h_2(a) \geq (1-p)h_2(d_0) + p h_2(1-d_1) = (1-p)h_2(d_0) + p h_2(d_1)$, where the strict step holds because $u = a * q$ is strictly closer to $\frac{1}{2}$ for $q \in (0, \frac{1}{2}]$ and the second step by concavity; so $L > 0$. If $a = \frac{1}{2}$, then $u = \frac{1}{2}$, $h_2(u) = 1$, and $L = 1 - (1-p)h_2(d_0) - p h_2(d_1) \geq 1 - h_2(D) > 0$ by concavity at the distortion. $\square$

Global optimality of the family-plus-tilt solution over all channels, including larger reproduction alphabets, has additionally been verified numerically by unconditional Lagrangian certificates, which assume no convexity.

IX. DISCUSSION

APPENDIX A

SINGLE-LETTERIZATION

For completeness, we justify the last inequalities in (11) and (12). For the rate coordinate,

$$\begin{aligned}
 I(X^n; \hat{X}^n) &= H(X^n) - H(X^n | \hat{X}^n) \\
 &= \sum_{i=1}^n H(X_i) - \sum_{i=1}^n H(X_i | X^{i-1}, \hat{X}^n) \\
 &\geq \sum_{i=1}^n \left[H(X_i) - H(X_i | \hat{X}_i) \right] \\
 &= \sum_{i=1}^n I(X_i; \hat{X}_i).
 \end{aligned} \tag{42}$$

For the work coordinate,

$$\begin{aligned}
 I(X^n; \hat{X}^n | S^n) &= H(X^n | S^n) - H(X^n | \hat{X}^n, S^n) \\
 &= \sum_{i=1}^n H(X_i | S_i) - \sum_{i=1}^n H(X_i | X^{i-1}, \hat{X}^n, S^n) \\
 &\geq \sum_{i=1}^n \left[H(X_i | S_i) - H(X_i | \hat{X}_i, S_i) \right] \\
 &= \sum_{i=1}^n I(X_i; \hat{X}_i | S_i).
 \end{aligned} \tag{43}$$

APPENDIX B

NUMERICAL VERIFICATION

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